Q1. **[9 Points]** Kalman Filter for Pose Estimation

a. [1 Points] Write out the symbolic Kalman Filter equations
b. [2 Point] Explain in your own words the matrices A, B, P, Q, R *and* K
c. [1 Points] What is the difference between Kalman Filter and Extended Kalman Filter?
d. [1 Point] How does the Kalman filter (or EKF) help with pose estimation in AR?

a) $\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$

$P_k^- = AP_{k-1}A^T + Q$

$K_k = P_k^- H^T (HP_k^- H^T + R)^{-1}$

$\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-)$

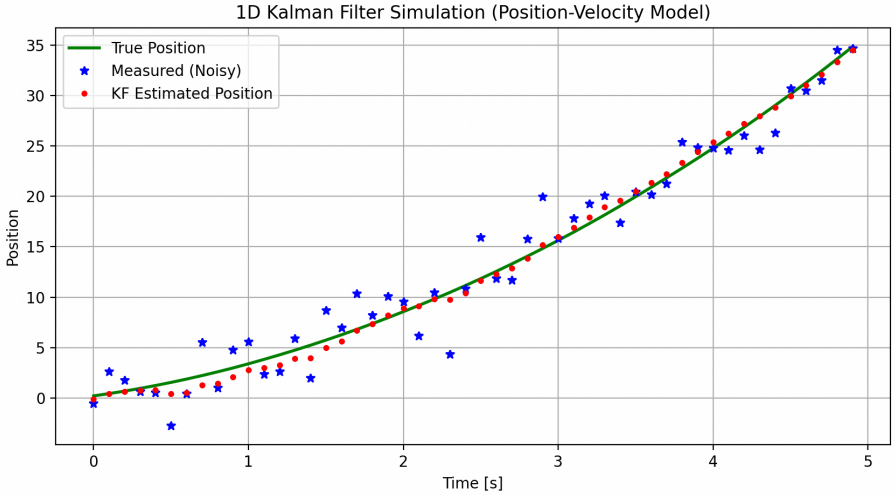
$P_k = (1 - K_k H)P_k^-$

b) The A matrix is often the identity matrix. It maps the next state from the previous state w/o any input. B maps the input, u, into its effect on the state. So, if there are no inputs, B is nonexistant. P is the covariance matrix for the state. It represents the uncertainty of the estimate. Q is the covariance of the process noise. It represents how much you trust the model. R is the covariance matrix for the measurements. It represents how much you trust the sensors. K is the Kalman gain. It balances the state prediction with how much to correct it based on sensor data.

c) The difference between the Kalman filter and the extended Kalman filter is that the EKF linearizes the process and/or system model whereas the KF assumes a linear system.

d) The KF/EKF helps w/ pose estimation by taking an initial prediction and uses sensor data to enhance that prediction. It is especially helpful in AR for position tracking with the HoloLens.

e)



f)

